

# Cost Analysis

2026-06-08

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## 1 Assumptions

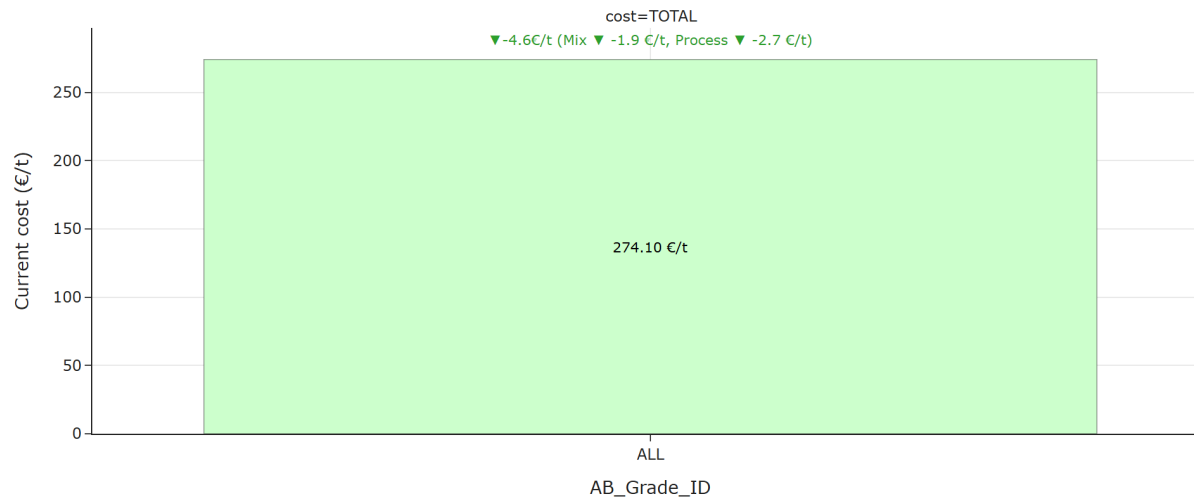
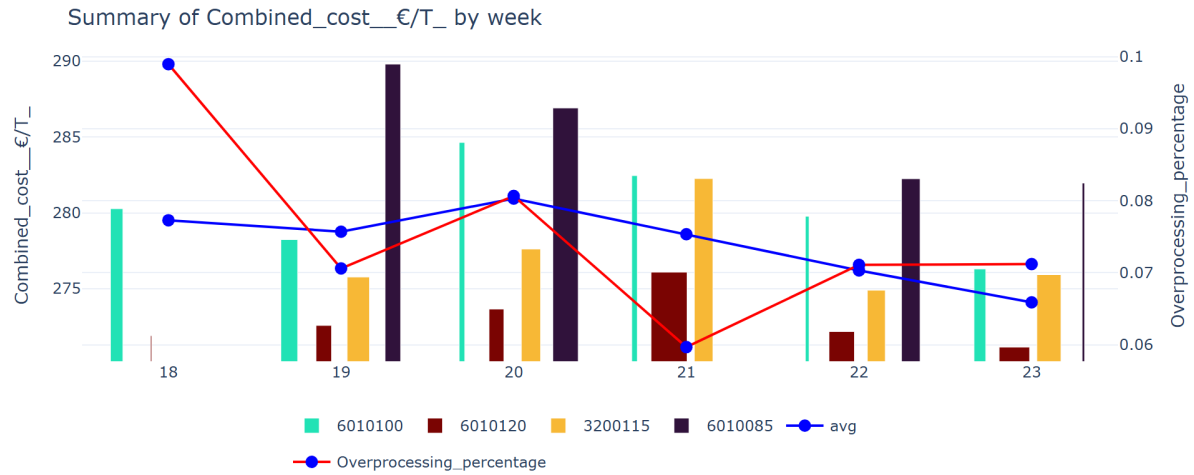
The aim of the analysis is to identify the most cost-effective production periods in months of data and recommend optimal operating windows for the corresponding key influencers that can be set as a benchmark for the weekly analysis.

- **Period Analyzed:** May 03, 2026 – June 08, 2026
- **Cost Components Included:** Fibre, steam, electricity, starch, and chemicals.
- **Grades Included:** Only grades 6010085, 6010100, 601020 and 3200115 are included in this week’s analysis.
- **Data Selection Criteria:** Only operating conditions where strength properties exceed specification thresholds were considered.
- **Naming Conventions Used:**
  - **Current:** Refers to the last week of the analyzed period.
  - **Historic:** Represents the average cost over the monthly interval previous to the “Current” period. Used as baseline.
  - **Best:** Corresponds to the cost associated with the optimal batch configuration.

## 2 Executive Summary

The following plot illustrates how total cost varies across different grades, enabling a comparison between the current configuration and the historical baseline.

The next plot displays how total cost evolves over time, with a breakdown by grade to show individual contributions.



For all considered grades, TOTAL cost saw a decrease of 4.57 €/t, resulting in 274.10 €/t.

Breaking down the overall cost change, -1.90 €/t was driven by a cheaper mix of grades, and -2.67 €/t was due to an improvement in the process.



For grade 3200115, TOTAL cost saw a decrease of 1.65 €/t, resulting in 275.93 €/t.

For grade 6010085, TOTAL cost saw a decrease of 4.24 €/t, resulting in 281.97 €/t.

For grade 6010100, TOTAL cost saw a decrease of 3.98 €/t, resulting in 276.31 €/t.

For grade 6010120, TOTAL cost saw a decrease of 2.84 €/t, resulting in 271.15 €/t.

Looking across individual grades, the strongest cost decrease was observed for grade 6010085 (-4.24 €/t). These movements stand out and may require further investigation.



Average change by component:

- **Steam\_€/T** decreased on average (-3.62 €/t).
- **Starch\_€/T** increased on average (+0.34 €/t).
- **Electricity\_€/T** increased on average (+0.15 €/t).
- **Fibre\_cost\_€/T** decreased on average (-0.12 €/t).
- **Chemicals\_€/T** increased on average (+0.07 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Starch\_€/T** (+0.34 €/t on average), with the highest increase in grade 3200115 (+1.77 €/t) and the largest average decrease comes from **Steam\_€/T** (-3.62 €/t on average), with the biggest drop in grade 6010085 (-4.61 €/t). These components stand out and may require further investigation.



Average change by steam:

- **Steam\_Predryers** decreased on average (-3.17 €/t).
- **Steam\_Whitewater** decreased on average (-0.67 €/t).
- **Steam\_Afterdryers** increased on average (+0.28 €/t).
- **Steam\_Starchkitchen** decreased on average (-0.04 €/t).
- **Steam\_Heatexchangers** decreased on average (-0.02 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Steam\_Afterdryers** (+0.28 €/t on average), with the highest increase in grade 3200115 (+0.91 €/t) and the largest average decrease comes from **Steam\_Predryers** (-3.17 €/t on average), with the biggest drop in grade 3200115 (-3.54 €/t). These components stand out and may require further investigation.

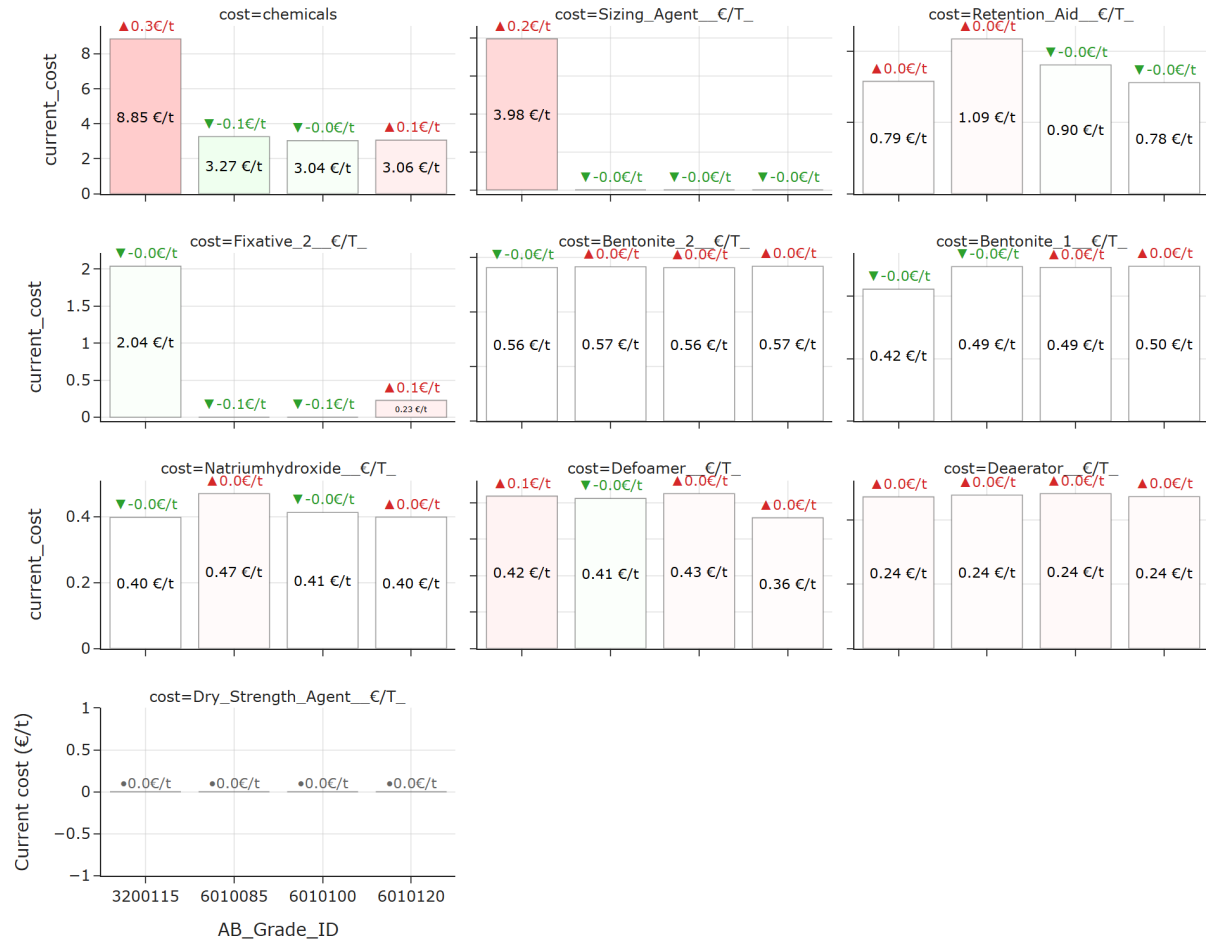


Average change by electricity:

- **Electricity\_Other** increased on average (+0.18 €/t).
- **Electricity\_Specific** decreased on average (-0.11 €/t).
- **Electricity\_Freqconverters** increased on average (+0.05 €/t).
- **Electricity\_Wiresection** increased on average (+0.05 €/t).
- **Electricity\_Press\_and\_Steambox** decreased on average (-0.01 €/t).
- **Electricity\_Predryer** increased on average (+0.01 €/t).
- **Electricity\_Afterdryer** decreased on average (-0.00 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Electricity\_Other** (+0.18 €/t on average), with the highest increase in grade 6010085 (+0.35 €/t) and the largest average decrease comes from **Electricity\_Specific** (-0.11 €/t on average),

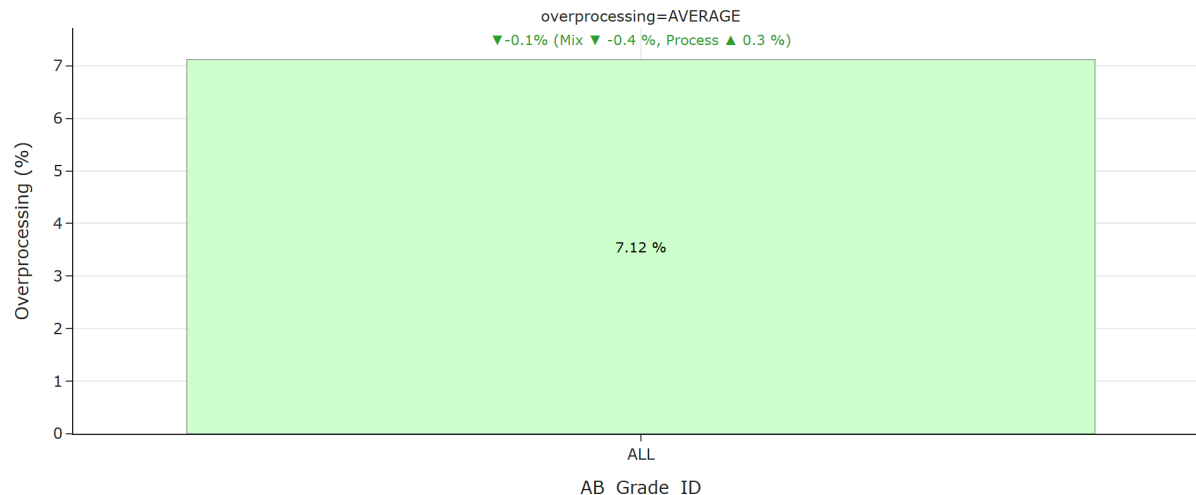
with the biggest drop in grade 6010100 (-0.37 €/t). These components stand out and may require further investigation.



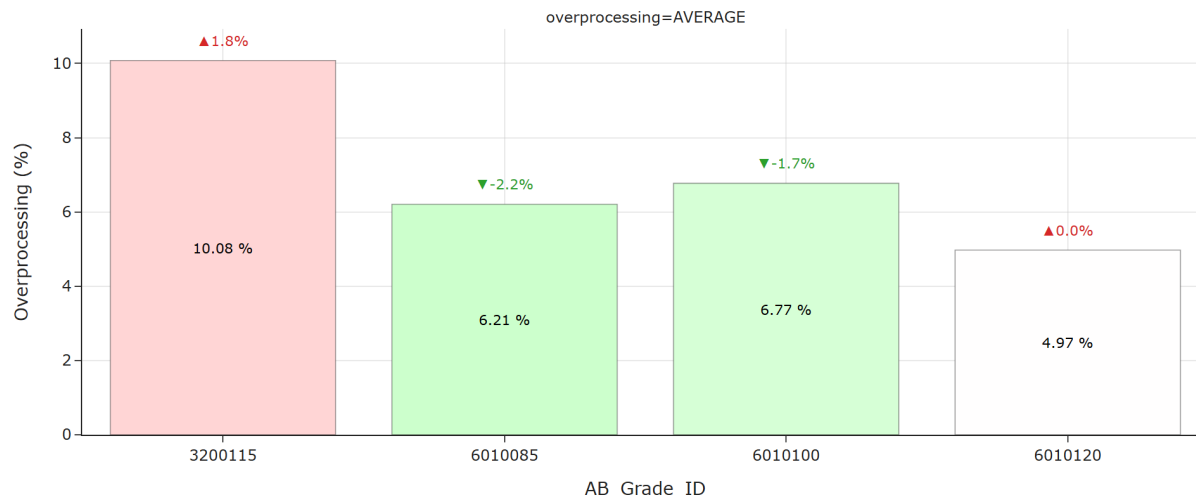
Average change by chemicals:

- **Sizing\_Agent\_€/T** increased on average (+0.25 €/t).
- **Fixative\_2\_€/T** increased on average (+0.03 €/t).
- **Deaerator\_€/T** increased on average (+0.03 €/t).
- **Defoamer\_€/T** increased on average (+0.03 €/t).
- **Retention\_Aid\_€/T** increased on average (+0.01 €/t).
- **Natriumhydroxide\_€/T** increased on average (+0.01 €/t).
- **Bentonite\_2\_€/T** increased on average (+0.00 €/t).
- **Bentonite\_1\_€/T** increased on average (+0.00 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Sizing\_Agent\_€/T** (+0.25 €/t on average), with the highest increase in grade 3200115 (+0.25 €/t). These components stand out and may require further investigation.



For all considered grades, TOTAL overprocessing saw a decrease of 0.09 %, resulting in 7.12 %.



For grade 3200115, TOTAL overprocessing saw an increase of 1.81 %, resulting in 10.08 %.

For grade 6010085, TOTAL overprocessing saw a decrease of 2.19 %, resulting in 6.21 %.

For grade 6010100, TOTAL overprocessing saw a decrease of 1.75 %, resulting in 6.77 %.

For grade 6010120, TOTAL overprocessing saw an increase of 0.03 %, resulting in 4.97 %.

Looking across individual grades, the strongest overprocessing increase was observed for grade 3200115 (+1.81 %) and the strongest overprocessing decrease was observed for grade 6010085 (-2.19 %). These movements stand out and may require further investigation.





Average change by component:

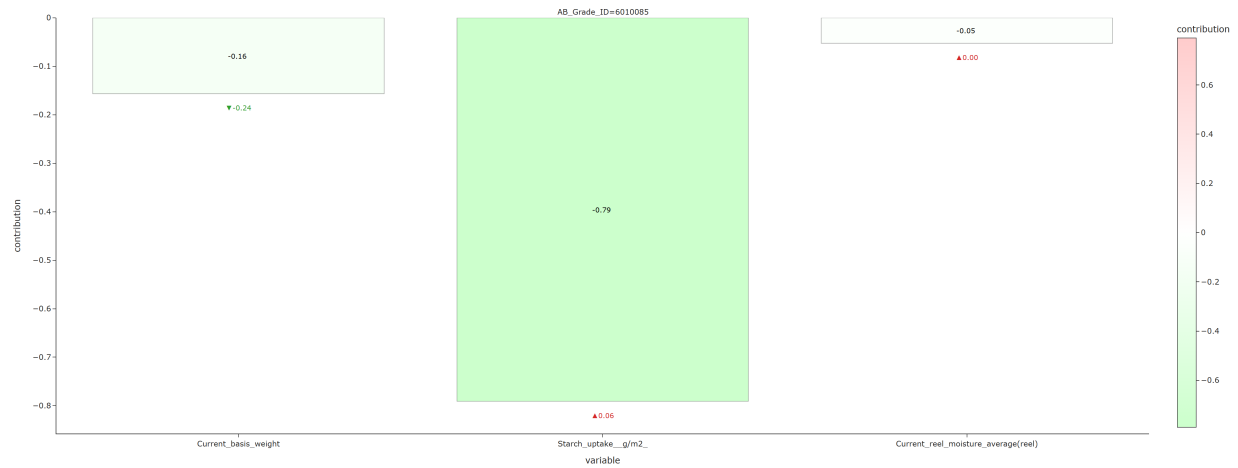
- **Burst** increased on average (+1.86 %).
- **SCT\_CD** decreased on average (-1.07 %).
- **CMT30** increased on average (+0.21 %).

A more detailed breakdown by component shows that the largest average increase comes from **Burst** (+1.86 % on average), with the highest increase in grade 3200115 (+1.86 %) and the largest average decrease comes from **SCT\_CD** (-1.07 % on average), with the biggest drop in grade 6010100 (-2.50 %). These components stand out and may require further investigation.

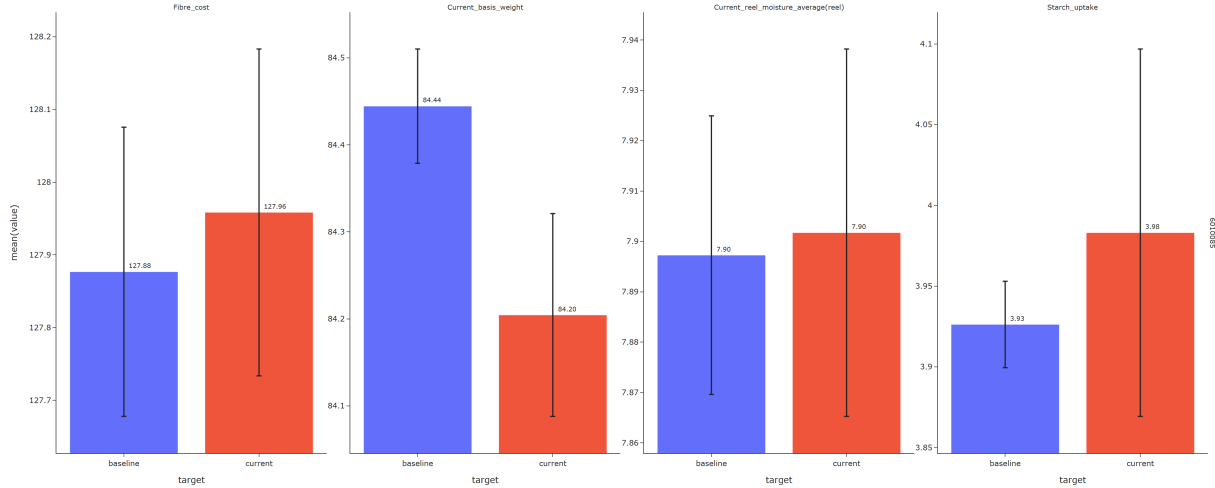
## 3 Breakdown by grades

### 3.1 6010085

#### 3.1.1 Fiber cost



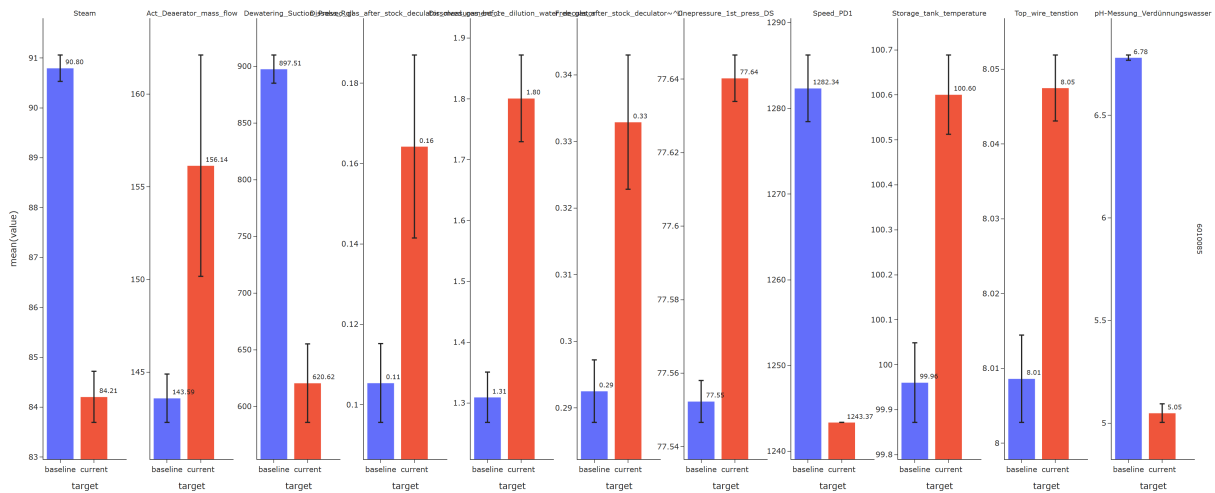
For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Starch uptake g/m2 increased (+0.06)



### 3.1.2 Steam cost



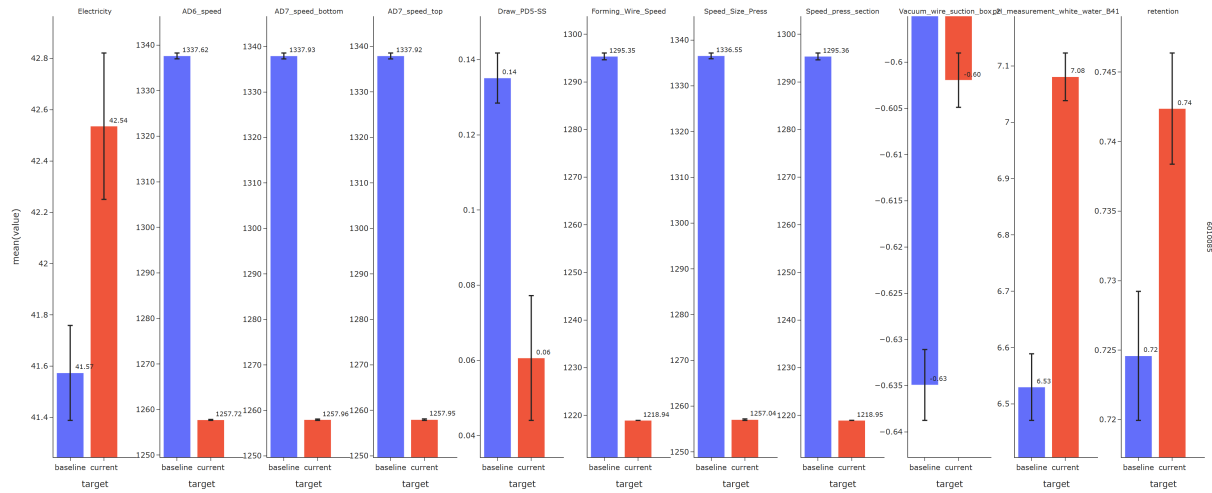
For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - PH-Messung Verdünnungswasser 2..12 pH decreased (-1.73) - Dissolved gas before dilution water decuator increased (+0.49) - Act Deaerator mass flow g/T increased (+12.55) - Top wire tension increased (+0.04) - Storage tank temperature increased (+0.64) - Linepressure 1st press DS bar increased (+0.09) Cost-increasing drivers: - Free gas after stock decuator~0 increased (+0.04)



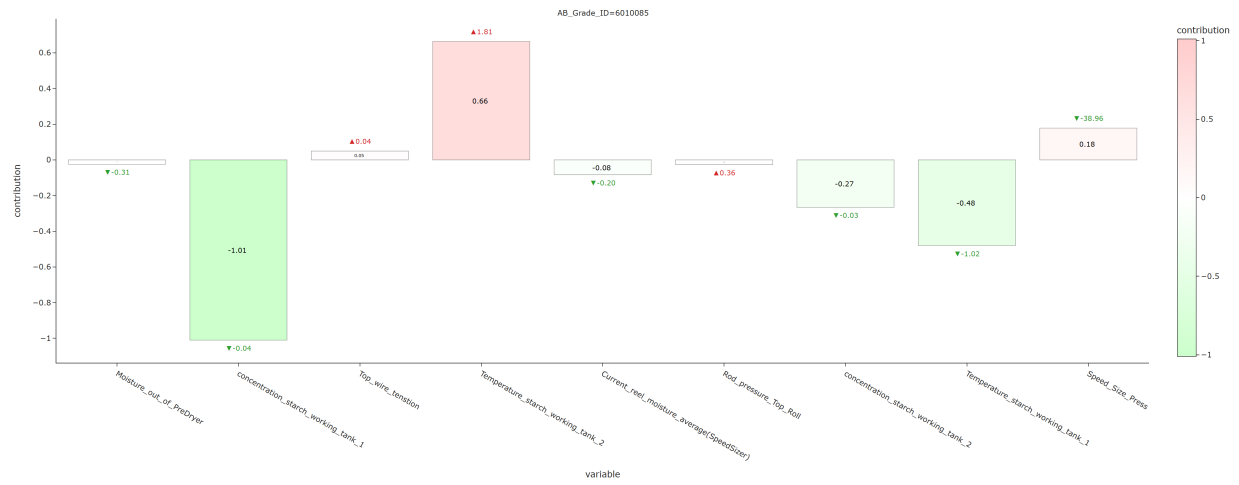
### 3.1.3 Electricity cost



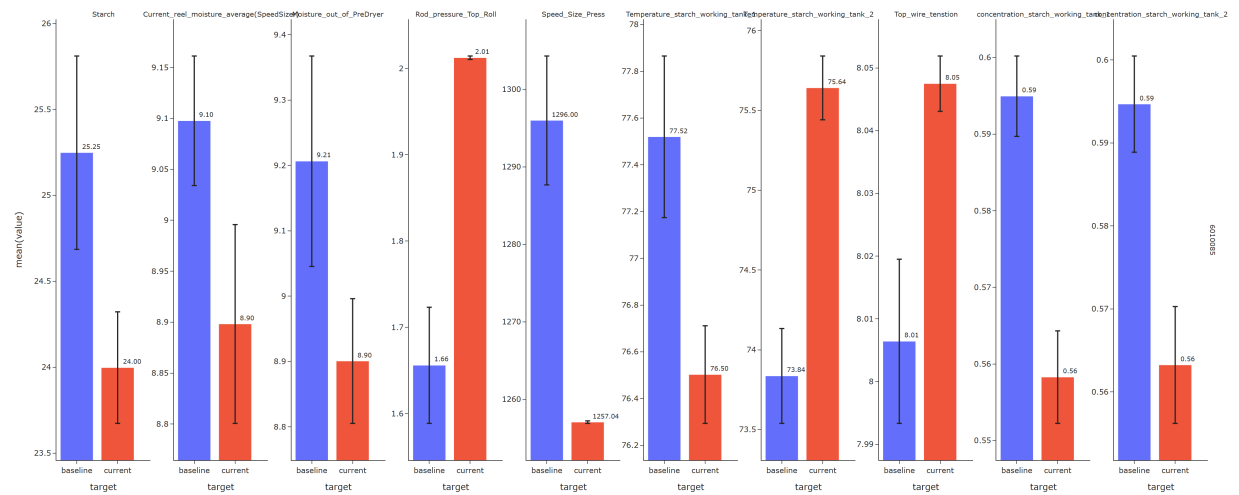
For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Vacuum wire suction box 2 increased (+0.03) - Retention increased (+0.02) Cost-increasing drivers: - PH measurement white water B41 increased (+0.55) - Draw PD5-SS decreased (-0.07) - Forming Wire Speed decreased (-76.41)



### 3.1.4 Starch cost

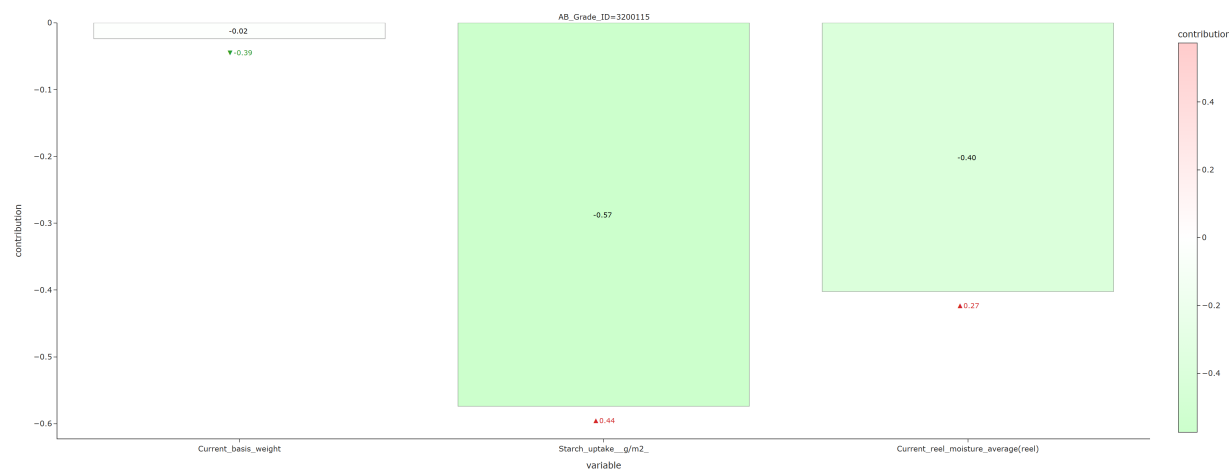


For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Concentration starch working tank 1 decreased (-0.04)  
Cost-increasing drivers: - Temperature starch working tank 2 increased (+1.81)

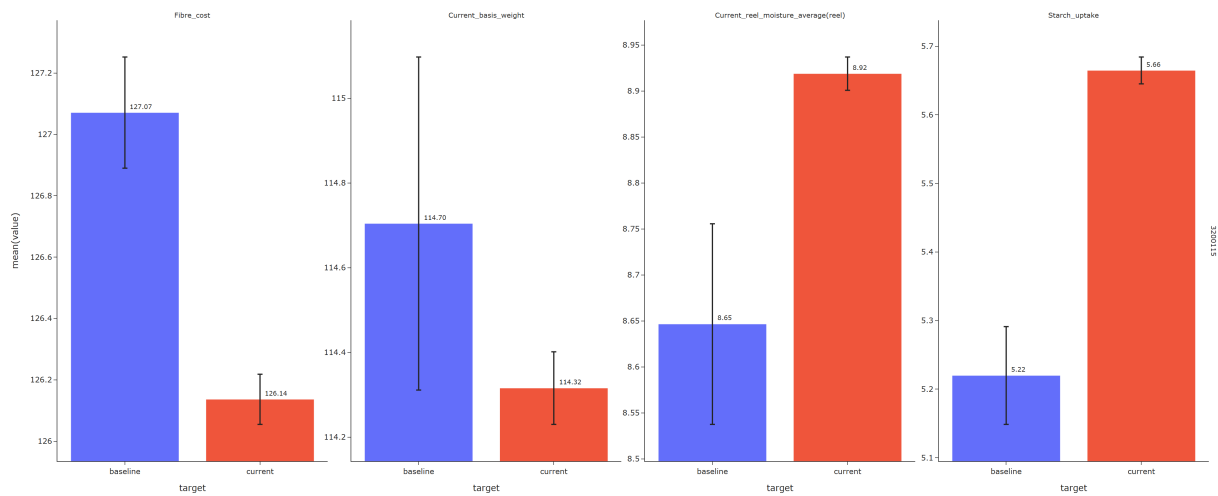


3.2 3200115

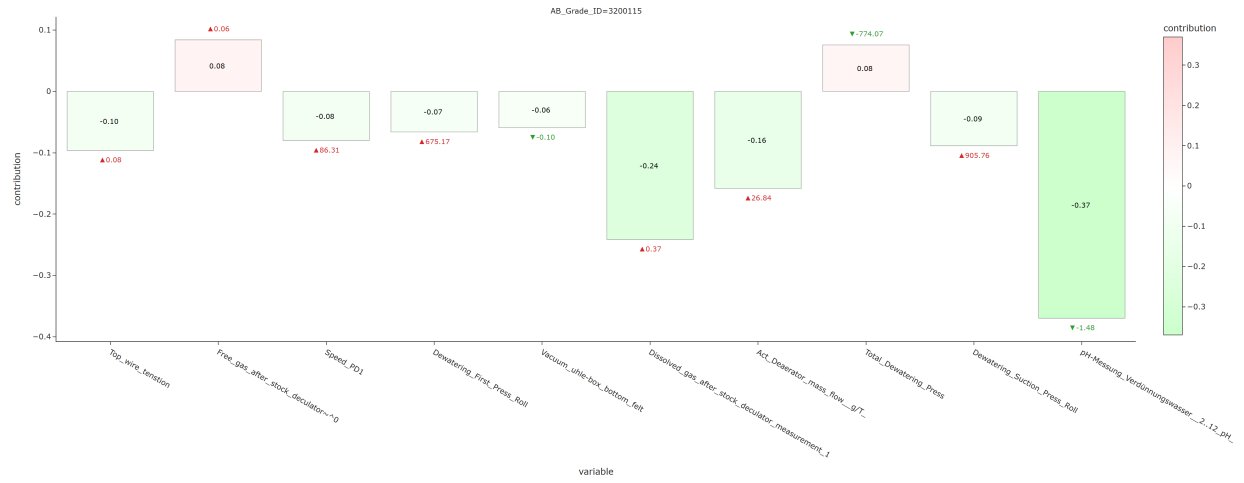
3.2.1 Fiber cost



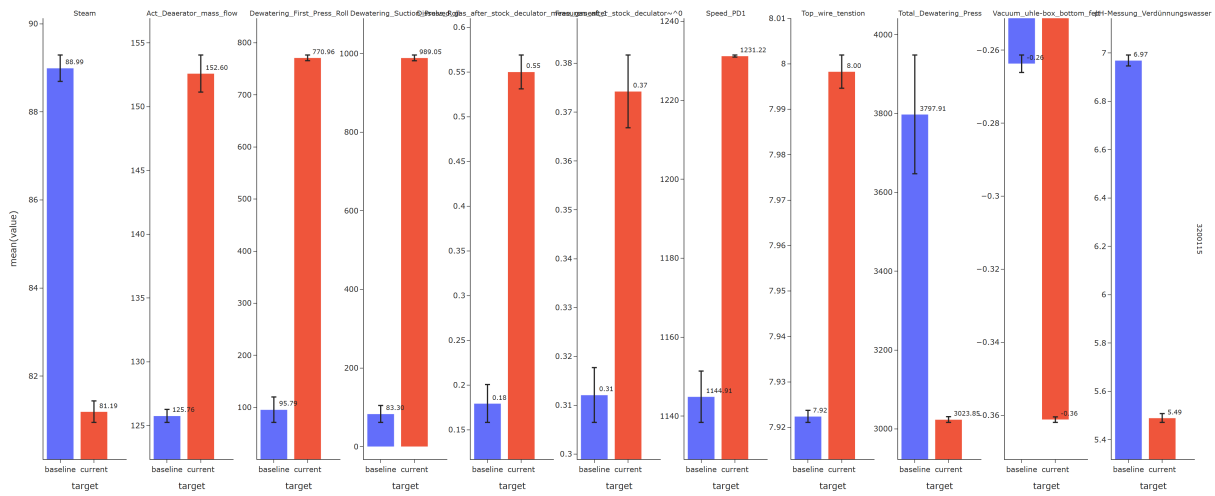
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Starch uptake g/m2 increased (+0.44)



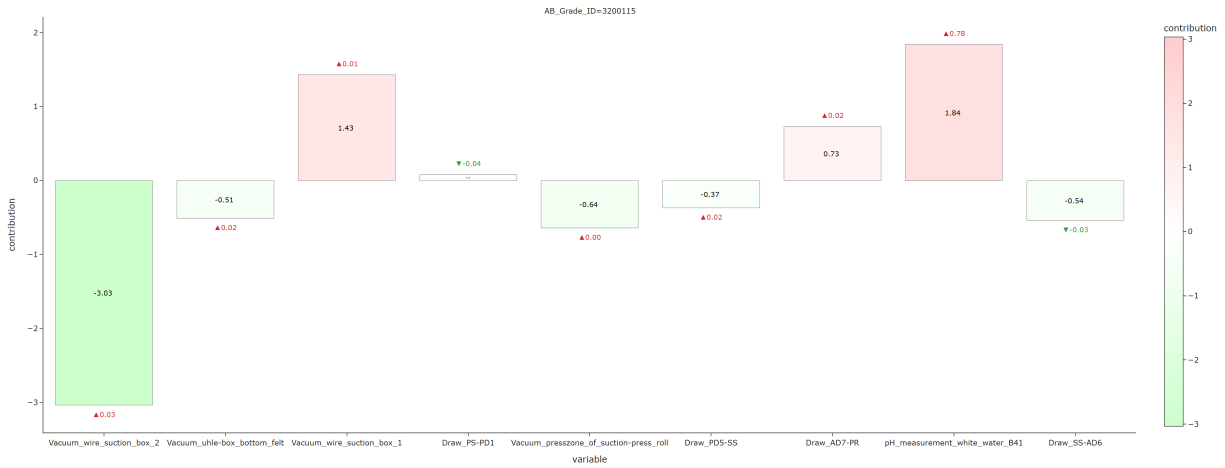
### 3.2.2 Steam cost



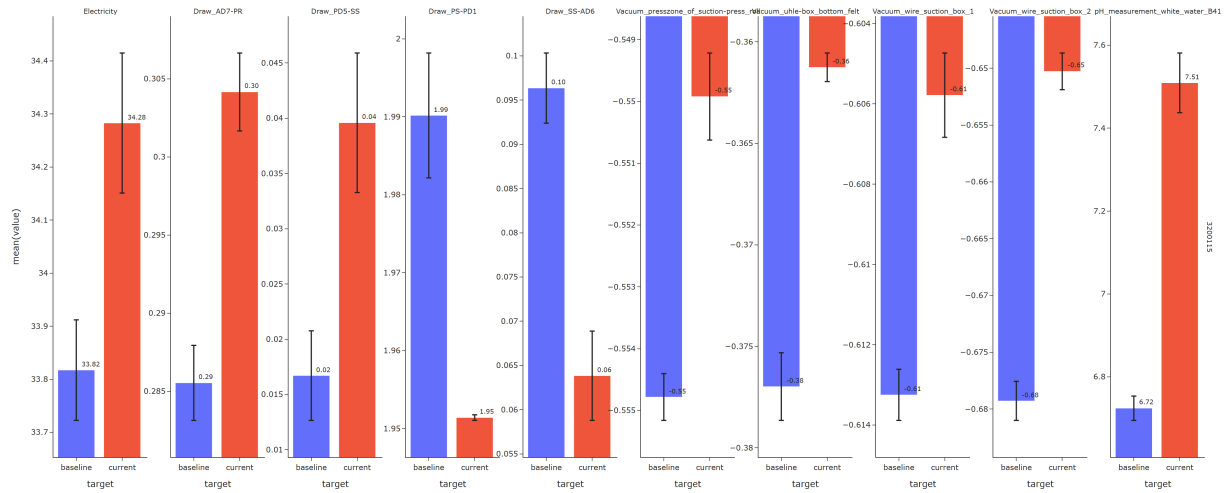
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - PH-Messung Verdünnungswasser 2..12 pH decreased (-1.48) - Dissolved gas after stock decudator measurement 1 increased (+0.37) - Act Deaeator mass flow g/T increased (+26.84) - Top wire tension increased (+0.08) - Dewatering Suction Press Roll increased (+905.76) Cost-increasing drivers: - Free gas after stock decudator~0 increased (+0.06)



### 3.2.3 Electricity cost

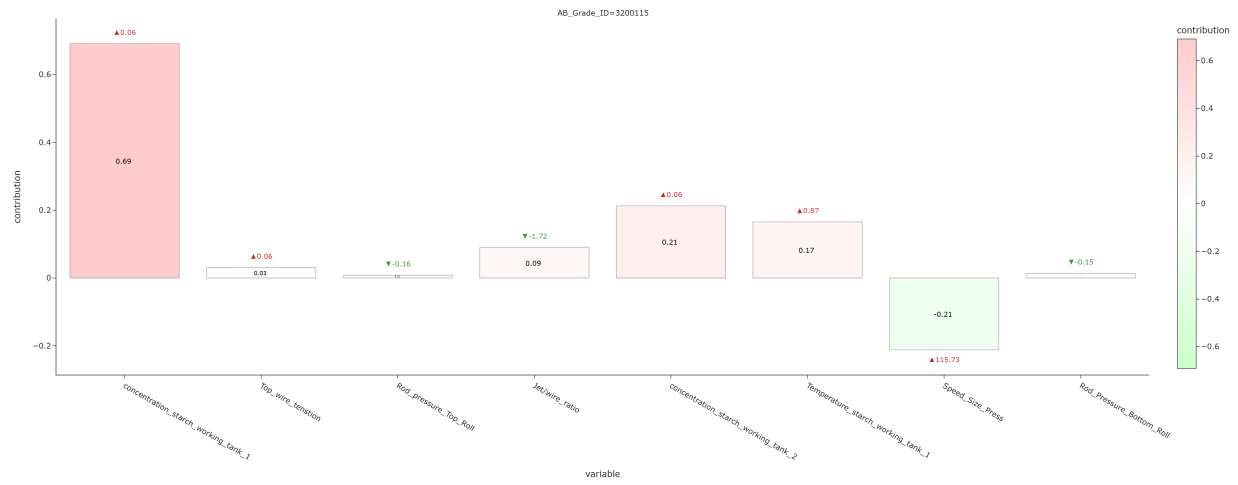


For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Vacuum wire suction box 2 increased (+0.03) Cost-increasing drivers: - PH measurement white water B41 increased (+0.78)

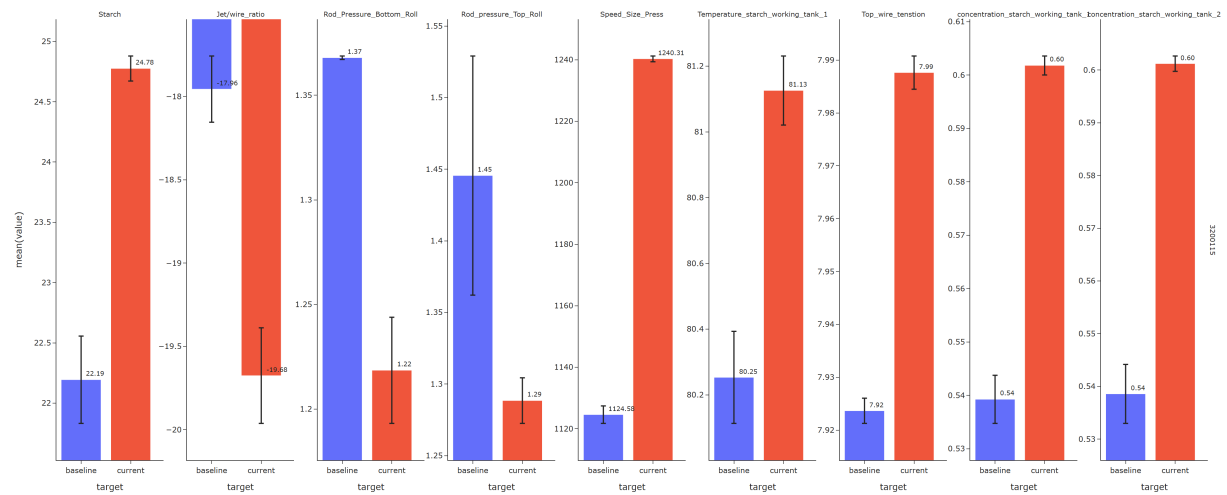




### 3.2.4 Starch cost

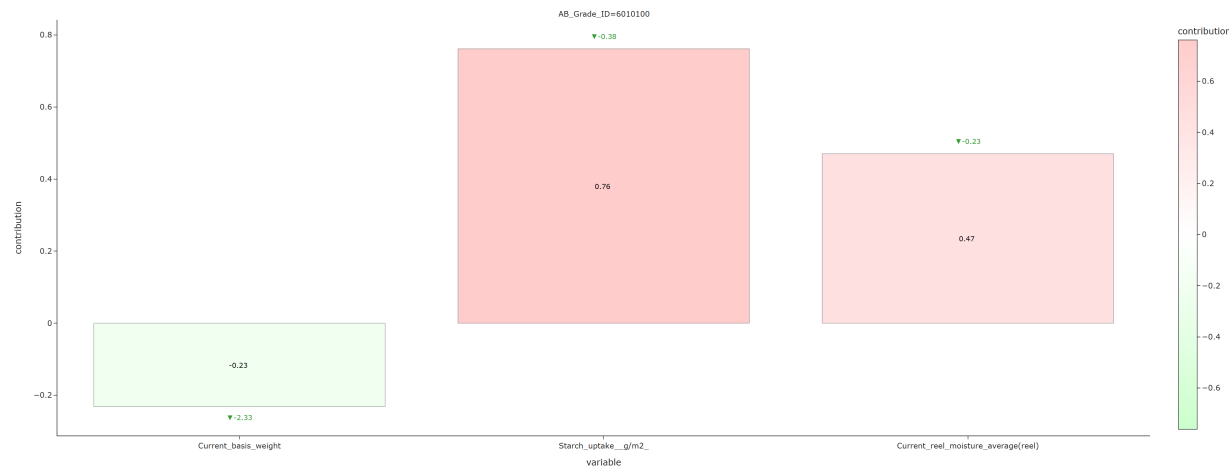


For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-increasing drivers: - Concentration starch working tank 1 increased (+0.06) - Concentration starch working tank 2 increased (+0.06)

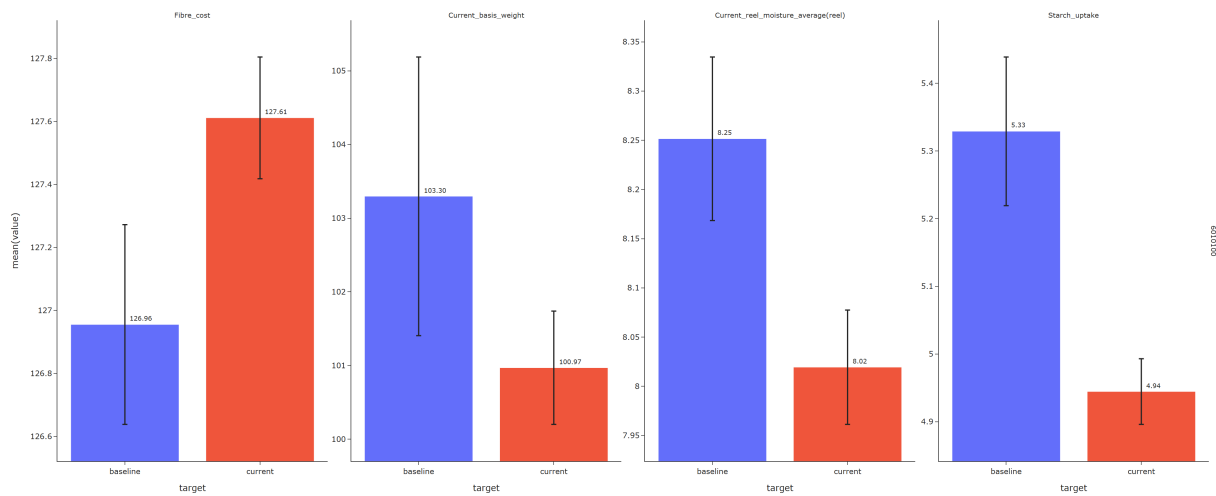


### 3.3 6010100

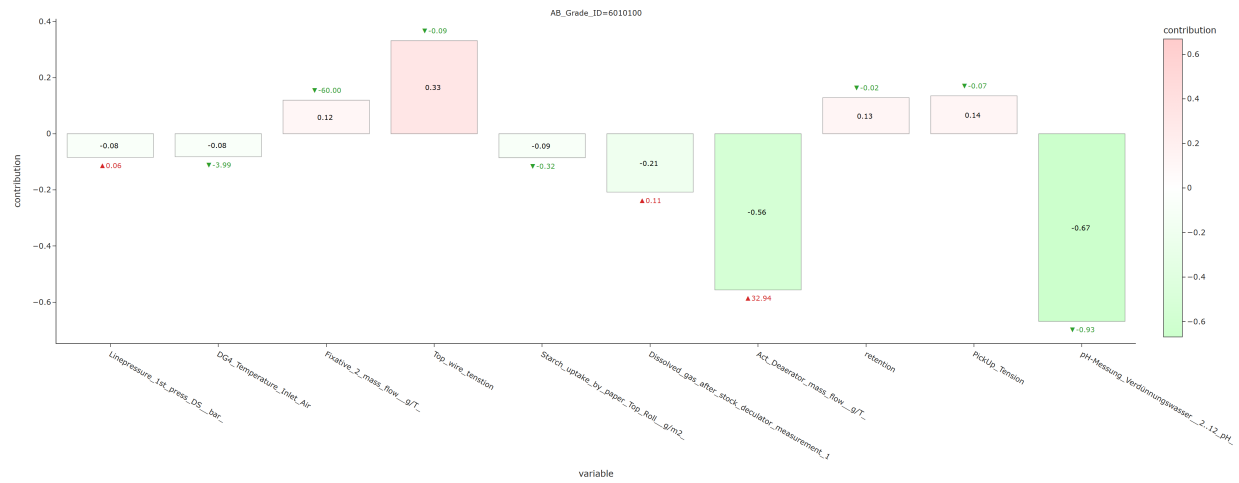
#### 3.3.1 Fiber cost



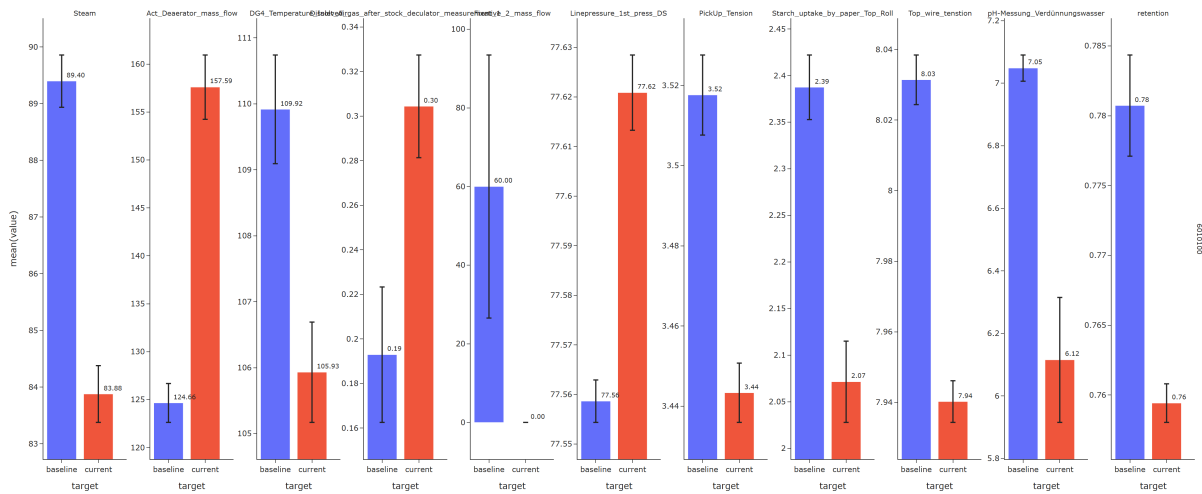
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-increasing drivers: - Starch uptake g/m2 decreased (-0.38)



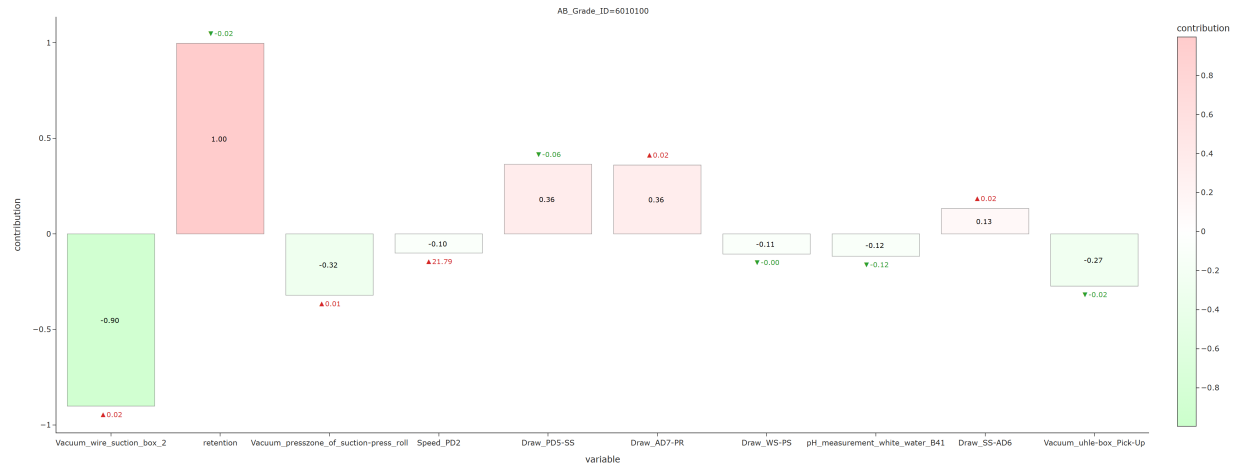
### 3.3.2 Steam cost



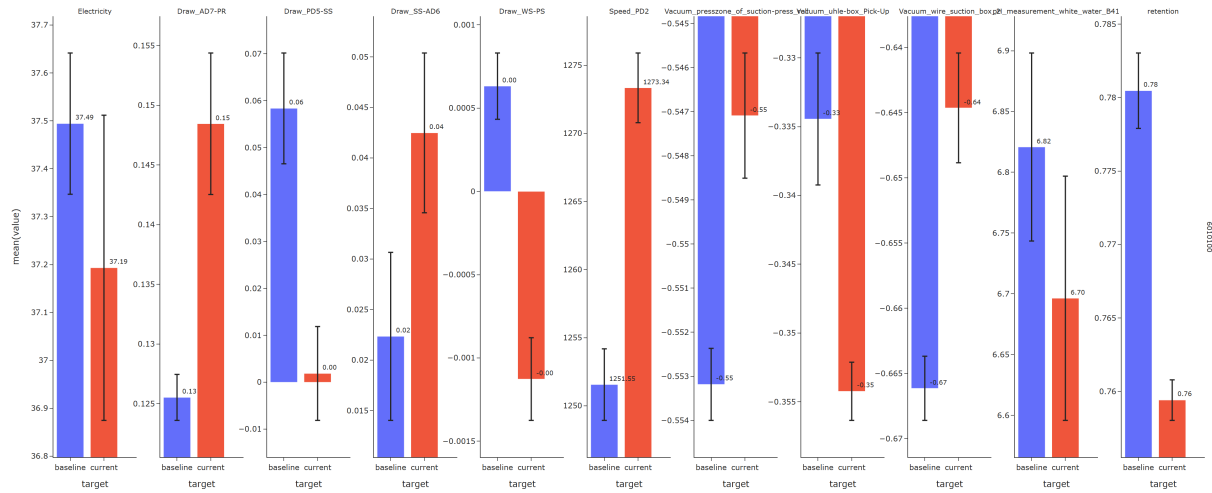
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - PH-Messung Verdünnungswasser 2..12 pH decreased (-0.93) - Act Deaeator mass flow g/T increased (+32.94) - Dissolved gas after stock decuator measurement 1 increased (+0.11) Cost-increasing drivers: - Top wire tension decreased (-0.09) - PickUp Tension decreased (-0.07) - Retention decreased (-0.02) - Fixative 2 mass flow g/T decreased (-60.00)



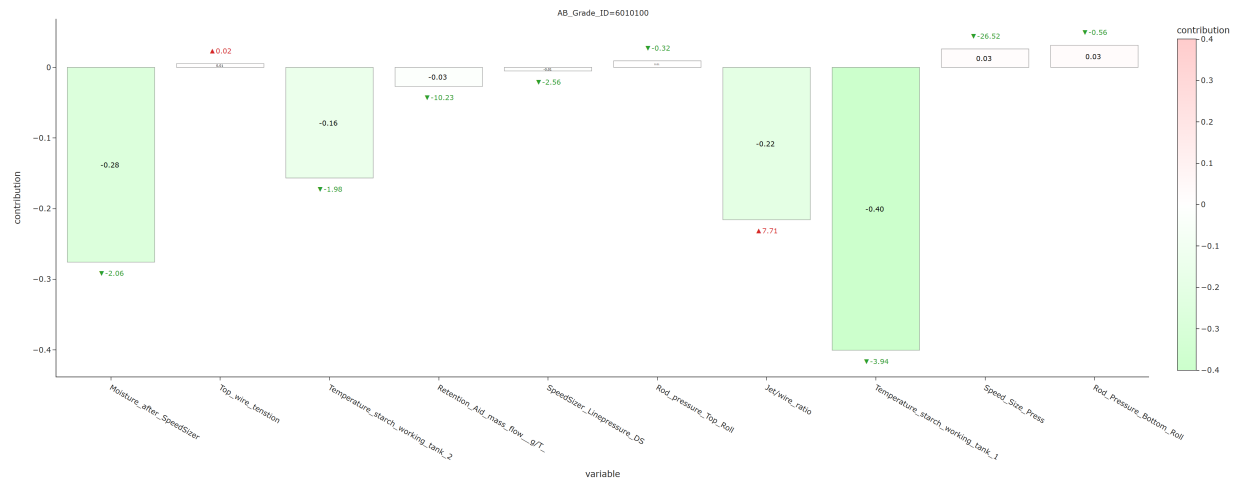
### 3.3.3 Electricity cost



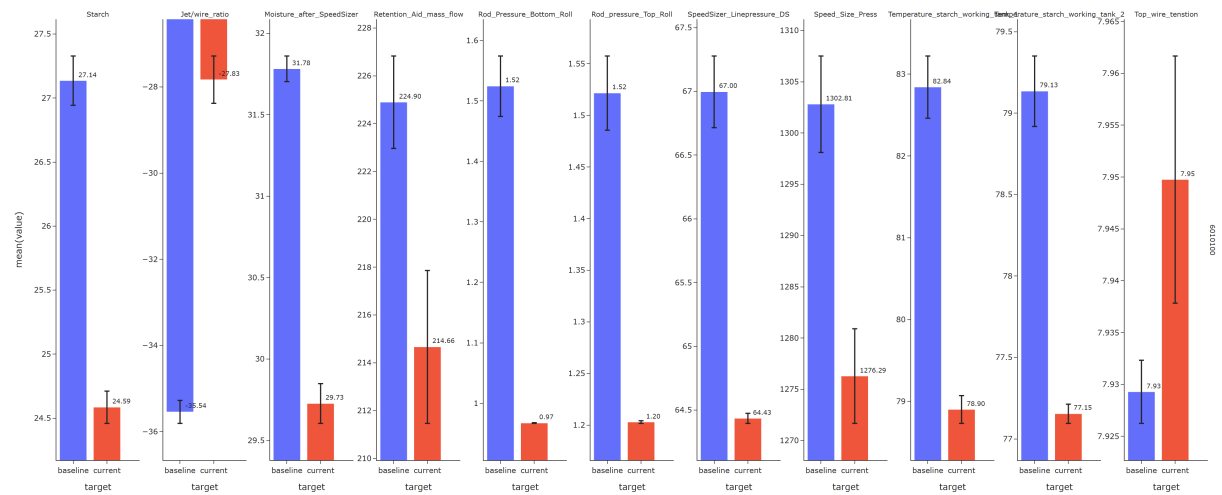
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Vacuum wire suction box 2 increased (+0.02) - Vacuum presszone of suction-press roll increased (+0.01) Cost-increasing drivers: - Retention decreased (-0.02) - Draw PD5-SS decreased (-0.06) - Draw AD7-PR increased (+0.02)



### 3.3.4 Starch cost

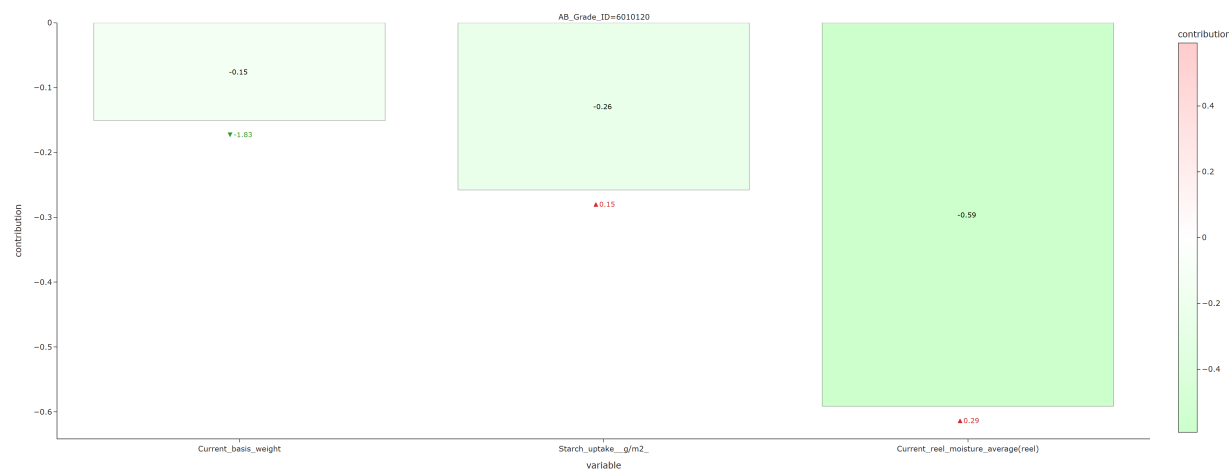


For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Temperature starch working tank 1 decreased (-3.94) - Moisture after SpeedSizer decreased (-2.06) - Jet/wire ratio increased (+7.71)

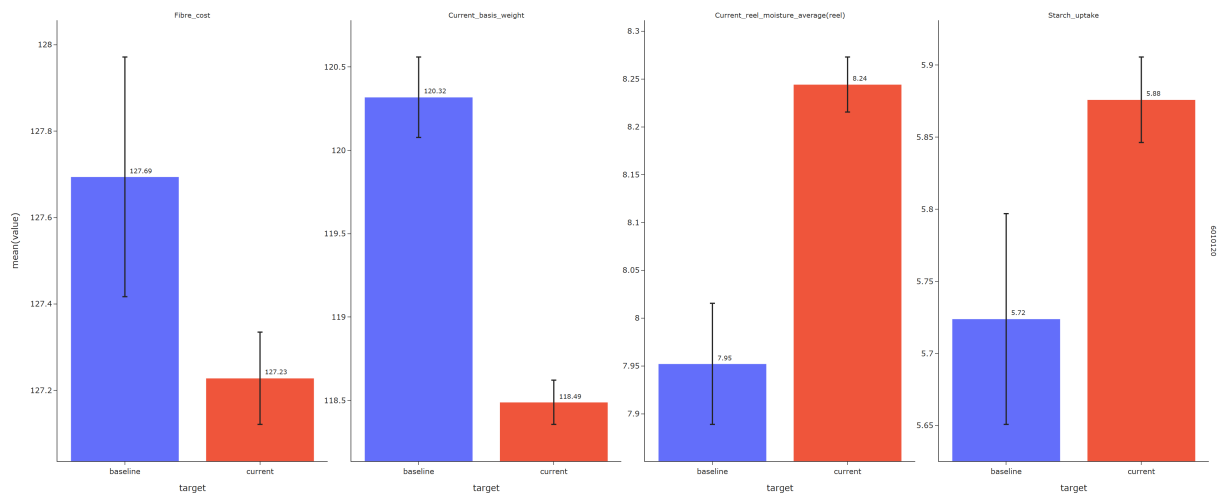


3.4 6010120

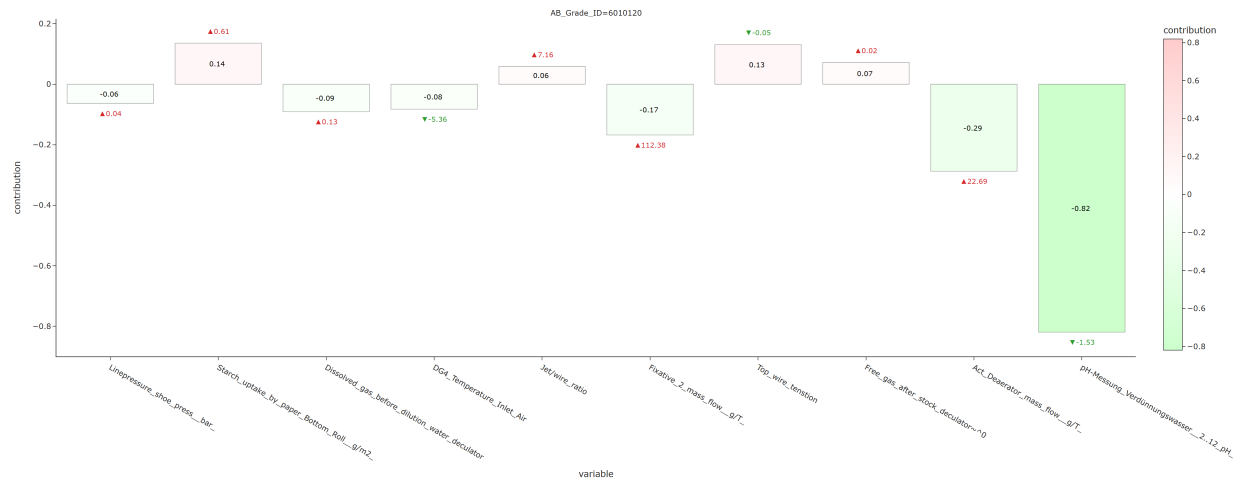
3.4.1 Fiber cost



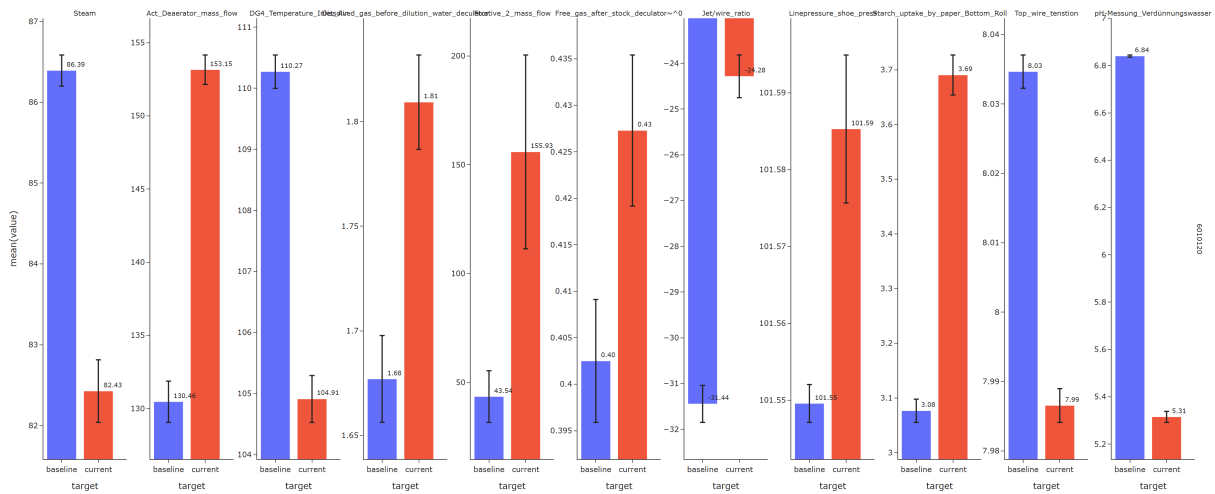
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Current reel moisture average(reel) increased (+0.29)



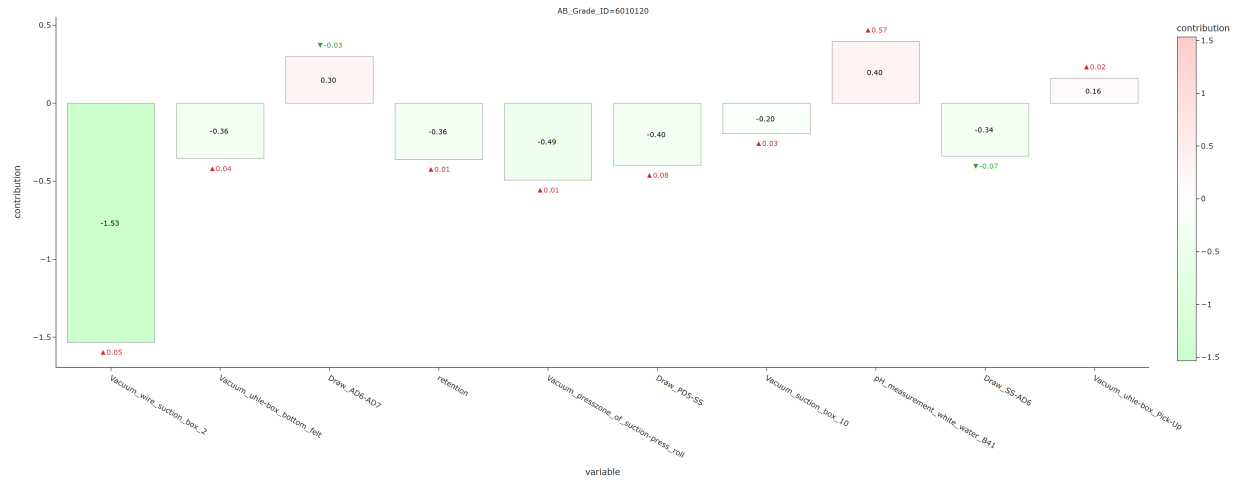
### 3.4.2 Steam cost



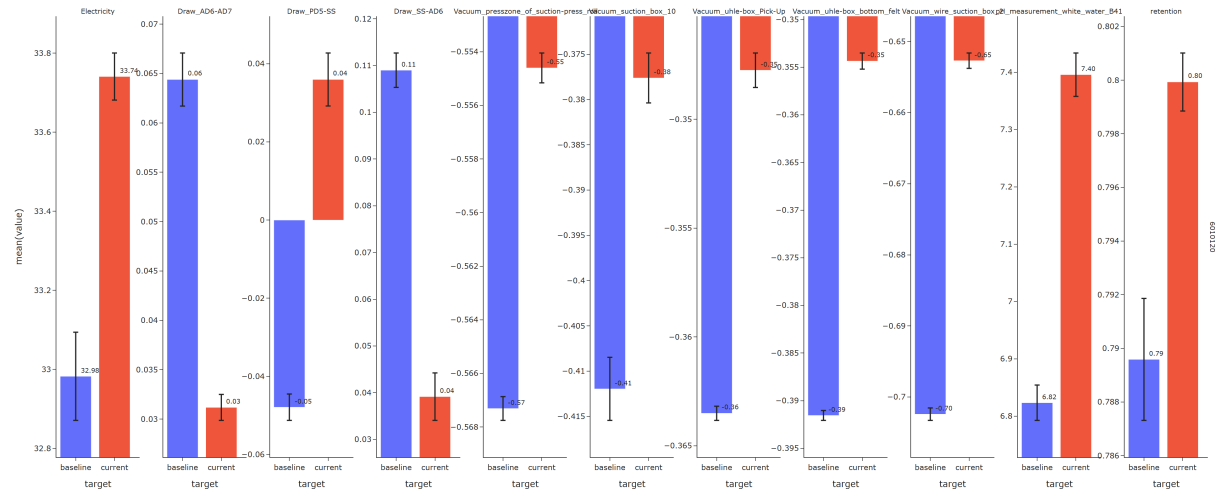
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - PH-Messung Verdünnungswasser 2..12 pH decreased (-1.53) - Act Deaerator mass flow g/T increased (+22.69) - Fixative 2 mass flow g/T increased (+112.38) Cost-increasing drivers: - Starch uptake by paper Bottom Roll g/m2 increased (+0.61) - Top wire tension decreased (-0.05)



### 3.4.3 Electricity cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Vacuum wire suction box 2 increased (+0.05) - Vacuum presszone of suction-press roll increased (+0.01) - Draw PD5-SS increased (+0.08) - Retention increased (+0.01) - Vacuum uhle-box bottom felt increased (+0.04) Cost-increasing drivers: - PH measurement white water B41 increased (+0.57)





### 3.4.4 Starch cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Jet/wire ratio increased (+5.98) Cost-increasing drivers: - Temperature starch working tank 1 increased (+3.78)

